APLICANDO TÉCNICAS DE DEVOPS – ENERGIA CERTA

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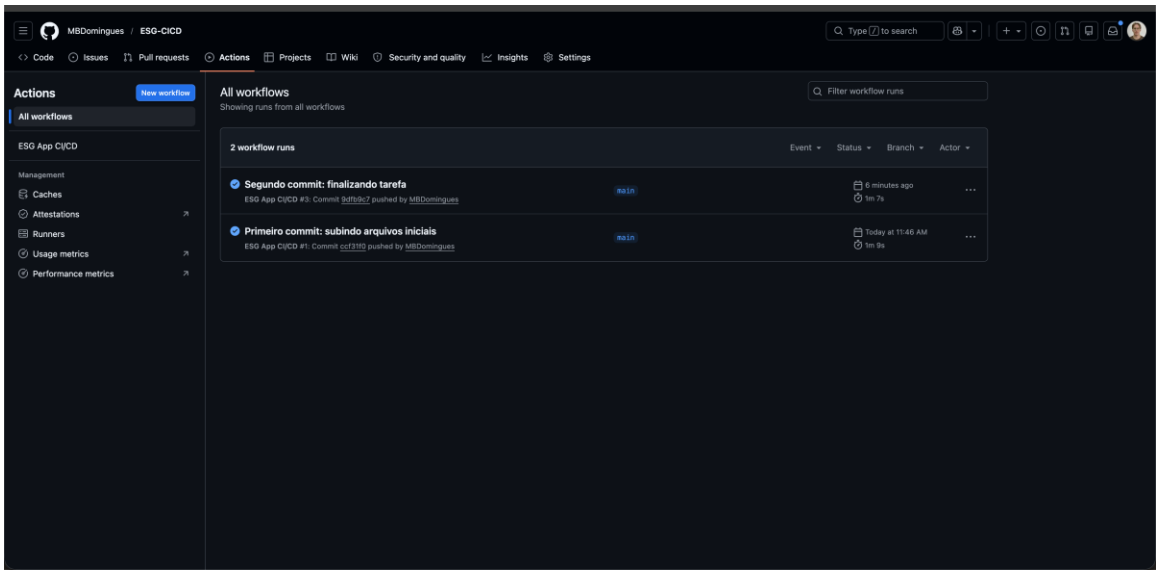
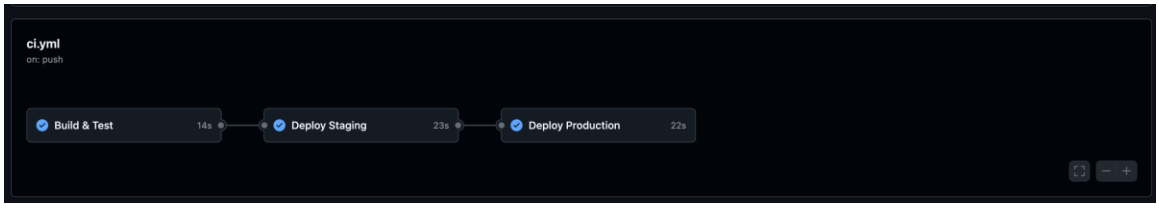
Descrição do Pipeline:

Ferramenta utilizada: GitHub Actions

Repositório: https://github.com/MBDomingues/ESG-CICD

O pipeline de CI/CD está configurado no arquivo .github/workflows/ci.yml e é automatizado em três etapas sequenciais:





1. Build & Test:

Instala as dependências do projeto com npm install, inicia o servidor Node.js em background e executa os testes automatizados (tests/api.test.cjs). Os testes validam o health check da API, listagem de dados das 5 collections, criação de novos registros, dashboard estatístico e tratamento de erros (404). Se qualquer teste falhar, o pipeline é interrompido e o deploy não prossegue.

```
Build & Test
succeeded 11 minutes ago in 14s
Search logs

Set up job                                                    1s
1  Current runner version: '2.333.1'
2  ▶ Runner Image Provisioner
9  ▶ Operating System
13 ▶ Runner Image
18 ▶ GITHUB_TOKEN Permissions
22 Secret source: Actions
23 Prepare workflow directory
24 Prepare all required actions
25 Getting action download info
26 Download action repository 'actions/checkout@v4' (SHA:34e114876b0b11c390a56381ad16ebd13914f8d5)
27 Download action repository 'actions/setup-node@v4' (SHA:49933ea5288caeca8642d1e84afbd3f7d6820020)
28 Complete job name: Build & Test

Checkout code                                                 0s
1  ▶ Run actions/checkout@v4
16 Syncing repository: MBDomingues/ESG-CICD
17 ▶ Getting Git version info
21 Temporarily overriding HOME='/home/runner/work/_temp/56e8238b-a8c4-4c4f-b93e-716cd3f00ced' before making global git config changes
22 Adding repository directory to the temporary git global config as a safe directory
23 /usr/bin/git config --global --add safe.directory /home/runner/work/ESG-CICD/ESG-CICD
24 Deleting the contents of '/home/runner/work/ESG-CICD/ESG-CICD'
25 ▶ Initializing the repository
42 ▶ Disabling automatic garbage collection
44 ▶ Setting up auth
52 ▶ Fetching the repository
56 ▶ Determining the checkout info
57 /usr/bin/git sparse-checkout disable
58 /usr/bin/git config --local --unset-all extensions.worktreeConfig
59 ▶ Checking out the ref
63 /usr/bin/git log -1 --format=%H
64 9dfb9c706714e263b246d310157ba291c79f005d
```

```
Setup Node.js                                                 2s
1  ▶ Run actions/setup-node@v4
8  Found in cache @ /opt/hostedtoolcache/node/20.20.2/x64
9  ▶ Environment details
13 /opt/hostedtoolcache/node/20.20.2/x64/bin/npm config get cache
14 /home/runner/.npm
15 Cache hit for: node-cache-Linux-x64-npm-ae9932306a0335ff159fc80feb8553d2f8978eba91f6dba14a1dd257d8a64441
16 Received 4690832 of 4690832 (100.0%), 9.5 MBs/sec
17 Cache Size: ~4 MB (4690832 B)
18 /usr/bin/tar -xf /home/runner/work/_temp/293b6c98-1e46-4c7c-821f-567b18e0cc4b/cache.tzst -P -C /home/runner/work/ESG-CICD/ESG-CICD --use-compress-program unzstd
19 Cache restored successfully
20 Cache restored from key: node-cache-Linux-x64-npm-ae9932306a0335ff159fc80feb8553d2f8978eba91f6dba14a1dd257d8a64441

Install dependencies                                          1s
1  ▶ Run npm install
4  npm warn deprecated prebuild-install@7.1.3: No longer maintained. Please contact the author of the relevant native addon; alternatives are available.
5
6  added 106 packages, and audited 107 packages in 970ms
7
8  24 packages are looking for funding
9    run `npm fund` for details
10
11 found 0 vulnerabilities
```

```
Run tests                                                                    8s
1  ▶Run # Start server in background
10 [seed] 10 fornecedores, 13 metricas, 10 alertas, 10 auditorias, 12 treinamentos inseridos.
11 [server] Database loaded with 10 fornecedores
12 [server] Energia Certa running on http://localhost:3000
13
14 > energia-certa-esg@1.0.0 test
15 > node tests/api.test.cjs
16
17
18 --- ESG API Tests ---
19
20 PASS Health endpoint returns ok
21 PASS List fornecedores returns array
22 PASS Create fornecedor
23 PASS List metricas
24 PASS List alertas
25 PASS List auditorias
26 PASS List treinamentos
27 PASS Dashboard returns stats
28 PASS Delete non-existent returns 404
29
30 --- Results: 9 passed, 0 failed ---
31

Post Setup Node.js                                                           0s
1 Post job cleanup.
2 Cache hit occurred on the primary key node-cache-Linux-x64-npm-ae9932306a0335ff159fc80feb8553d2f8978eba91f6dba14a1dd257d8a64441, not saving cache.

Post Checkout code                                                           0s
1 Post job cleanup.
2 /usr/bin/git version
3 git version 2.53.0
4 Temporarily overriding HOME='/home/runner/work/_temp/b803f2a1-a747-427f-bb0a-2121b0bee303' before making global git config changes
5 Adding repository directory to the temporary git global config as a safe directory
6 /usr/bin/git config --global --add safe.directory /home/runner/work/ESG-CICD/ESG-CICD
7 /usr/bin/git config --local --name-only --get-regexp core\.sshCommand
8 /usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'core\.sshCommand' && git config --local --unset-all 'core.sshCommand' || :"
9 /usr/bin/git config --local --name-only --get-regexp http\.https\:\/\/github\.com\/\.extraheader
10 http.https://github.com/.extraheader
11 /usr/bin/git config --local --unset-all http.https://github.com/.extraheader
12 /usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'http\.https\:\/\/github\.com\/\.extraheader' && git config --local --unset-all
   'http.https://github.com/.extraheader' || :"
13 /usr/bin/git config --local --name-only --get-regexp ^includeIf\.gitdir:
14 /usr/bin/git submodule foreach --recursive git config --local --show-origin --name-only --get-regexp remote.origin.url

Complete job                                                                 0s
1 Cleaning up orphan processes
2 Terminate orphan process: pid (2290) (node)
```

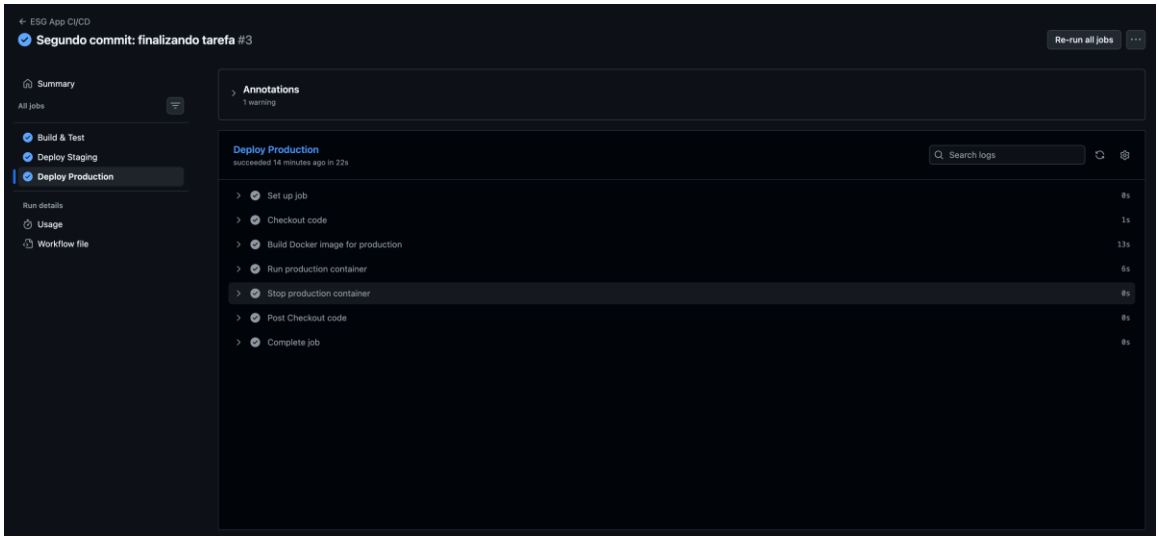
2. Deploy Staging:

Após a aprovação dos testes, é gerada uma imagem Docker com a tag energia-certa:v{run_number}-staging. O container é iniciado na porta 3001 com NODE_ENV=staging. Um health check via curl valida se a aplicação está respondendo.

```
← ESG App CI/CD
Segundo commit: finalizando tarefa #3                                  Re-run all jobs

Summary
All jobs
Build & Test
Deploy Staging
Deploy Production
Run details
Usage
Workflow file

Annotations
1 warning

Deploy Staging
succeeded 13 minutes ago in 23s                                        Search logs
Set up job                                                             0s
Checkout code                                                          1s
Setup Node.js                                                          0s
Build Docker image for staging                                         14s
Run staging container                                                  5s
Stop staging container                                                 0s
Post Setup Node.js                                                     0s
Post Checkout code                                                     1s
Complete job                                                           0s
```

3. Deploy Production:

Uma nova imagem Docker é gerada para produção (energia-certa:v{run_number}), iniciada na porta 3000 com NODE_ENV=production, e validada via health check.



Docker:

Arquitetura utilizando Docker Compose com ambientes isolados de produção e staging, volumes persistentes e rede bridge dedicada.

Dockerfile e Estratégias:

```
FROM node:20-alpine

  RUN apk add --no-cache python3 make g++

  WORKDIR /app

  COPY package.json .

  RUN npm install --production

  COPY . .

  RUN mkdir -p /app/data

  EXPOSE 3000

  ENV NODE_ENV=production

  ENV PORT=3000

  CMD ["node", "src/server.cjs"]
```

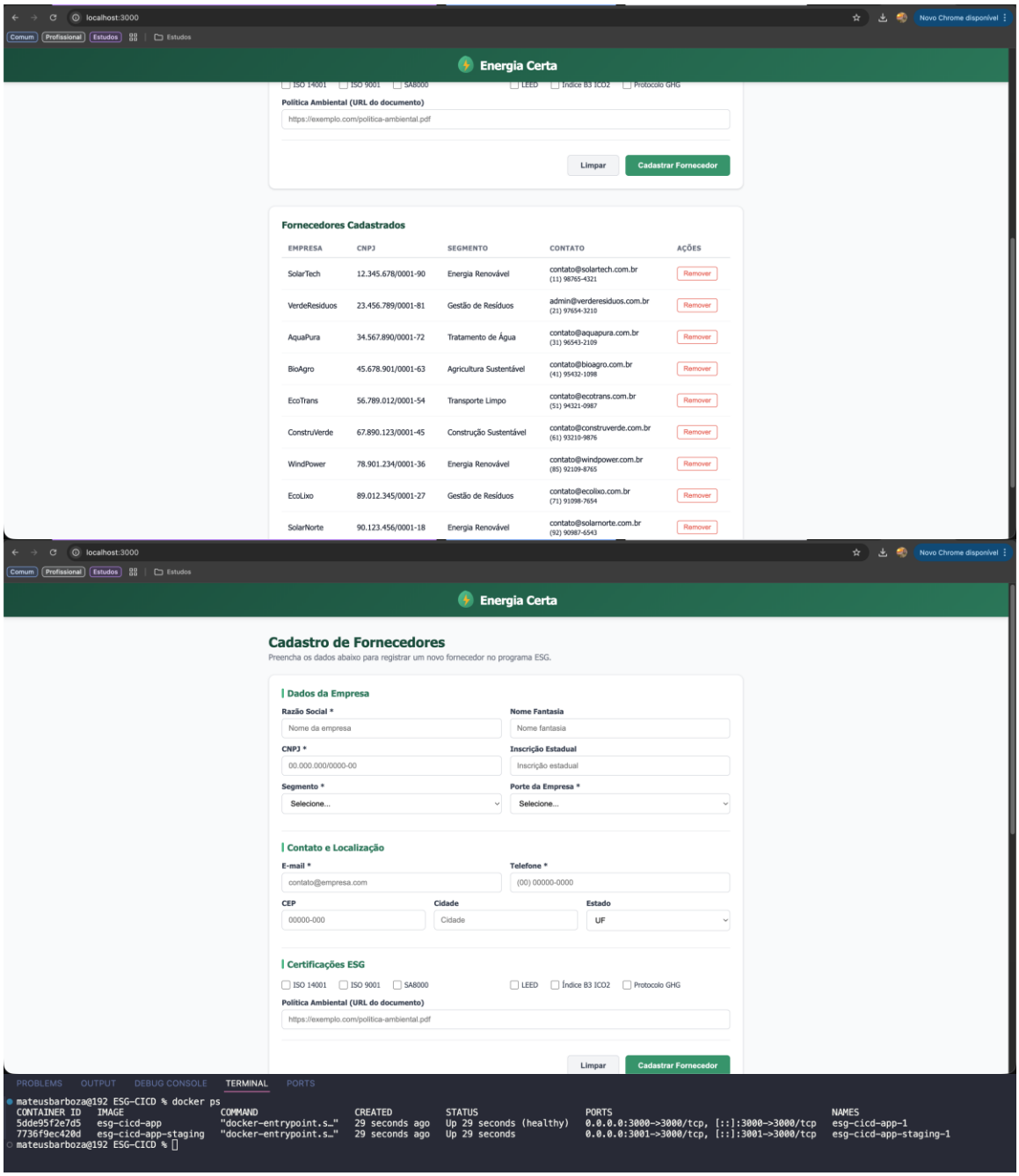
Uso de imagem node:20-alpine, instalação de dependências nativas para better-sqlite3, otimização de camadas e exclusão de dependências de desenvolvimento.

Comandos usados para rodar localmente:

```
  # Build da imagem

  docker build -t energia-certa:latest .

  # Subir ambos os ambientes (staging + produção)

  docker compose up --build

  # Verificar logs

  docker compose logs -f

  # Parar e remover containers

  docker compose down
```

Funcionamento e descrição do projeto:

O projeto foi baseado em temas de ESG anteriores e se trata de um crud basico de fornecedores.



Desafios Encontrados

1. Compilação do better-sqlite3

2. Foreign Keys no SQLite

3. Modelo híbrido NoSQL

4. Separação de ambientes

5. Testes sem frameworks externos